

8 Silane, SiH_4 , exists as a gas at standard temperature and pressure. Hess' Law can be used to calculate the average Si–H bond energy in gaseous SiH_4 . Which information is needed to perform the calculation?

A $\Delta H^\ominus_{\text{atomisation}}(\text{Si})$, $\Delta H^\ominus_{\text{combustion}}(\text{H}_2)$, $\Delta H^\ominus_{\text{formation}}(\text{SiH}_4)$

B $\Delta H^\ominus_{\text{atomisation}}(\text{Si})$, $\Delta H^\ominus_{\text{atomisation}}(\text{H})$, $\Delta H^\ominus_{\text{formation}}(\text{SiH}_4)$

C $\Delta H^\ominus_{\text{atomisation}}(\text{H})$, $\Delta H^\ominus_{\text{combustion}}(\text{Si})$, $\Delta H^\ominus_{\text{combustion}}(\text{SiH}_4)$

D $\Delta H^\ominus_{\text{combustion}}(\text{Si})$, $\Delta H^\ominus_{\text{combustion}}(\text{H}_2)$, $\Delta H^\ominus_{\text{formation}}(\text{SiH}_4)$